

8. The picture shows a label from a packet of pasta.

Typical values (as consumed)	per 100g
Energy	658kJ
	155Kcal
Fat	0.6g
of which saturates	0.1g
Carbohydrate	30.3g
of which sugars	0.4g
Fibre	3.4g
Protein	5.5g
Salt	trace

(a) From the label above, identify the nutrient that provides bulk in the diet. [1]

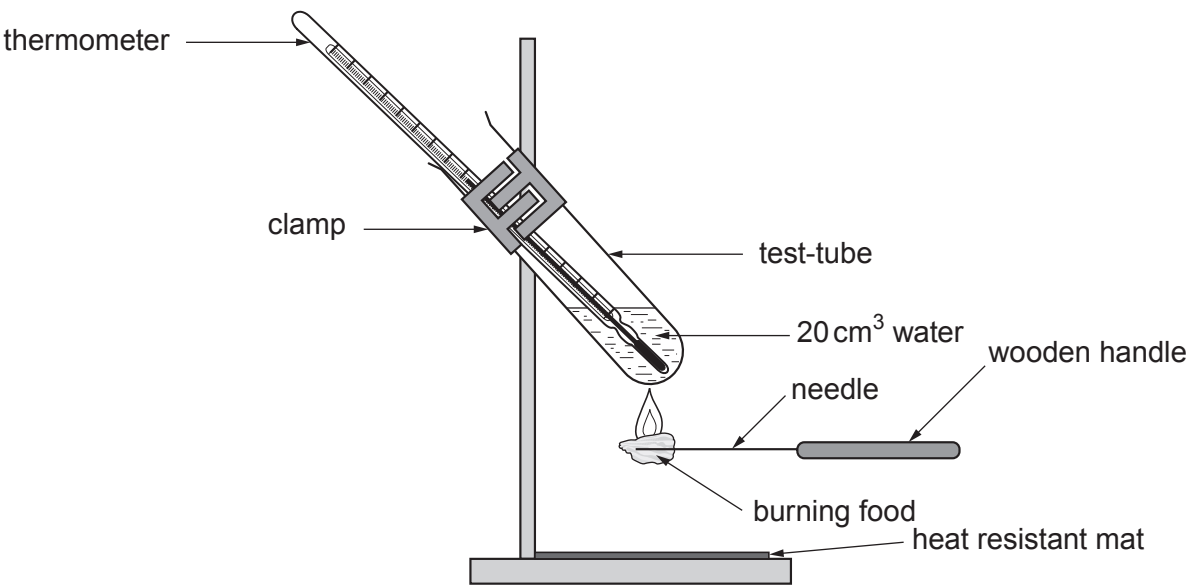
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(b) State why the government recommends that added sugars should not make up more than 5% of the energy obtained from food and drink each day. [1]

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Students compared the energy content of crisps and pasta using the following apparatus.



The energy content of food can be calculated using the following formula.

$$\text{Energy (J/g)} = \frac{(4.2 \times \text{volume of water} \times \text{change in temperature})}{\text{mass of food}}$$

The following results were obtained.

Table 1

Food	Mass of food (g)	Temperature of water in test tube (°C)		Change in temperature (°C)
		Initial	Final	
Crisp	0.7	20	59	39
Pasta	1.3	20	36	16



Examiner
only

The table below shows the energy values for the crisps.

Table 2

Food	Energy calculated from the experiment (J/g)	Energy calculated from the experiment (kJ/g)	Energy recorded on the food label (kJ/g)
Crisp	4680	4.680	22.788
Pasta			

- (c) Complete **Table 2** to show the energy values for pasta. You will need to refer to **Table 1** opposite and the food label on page 19. [4]

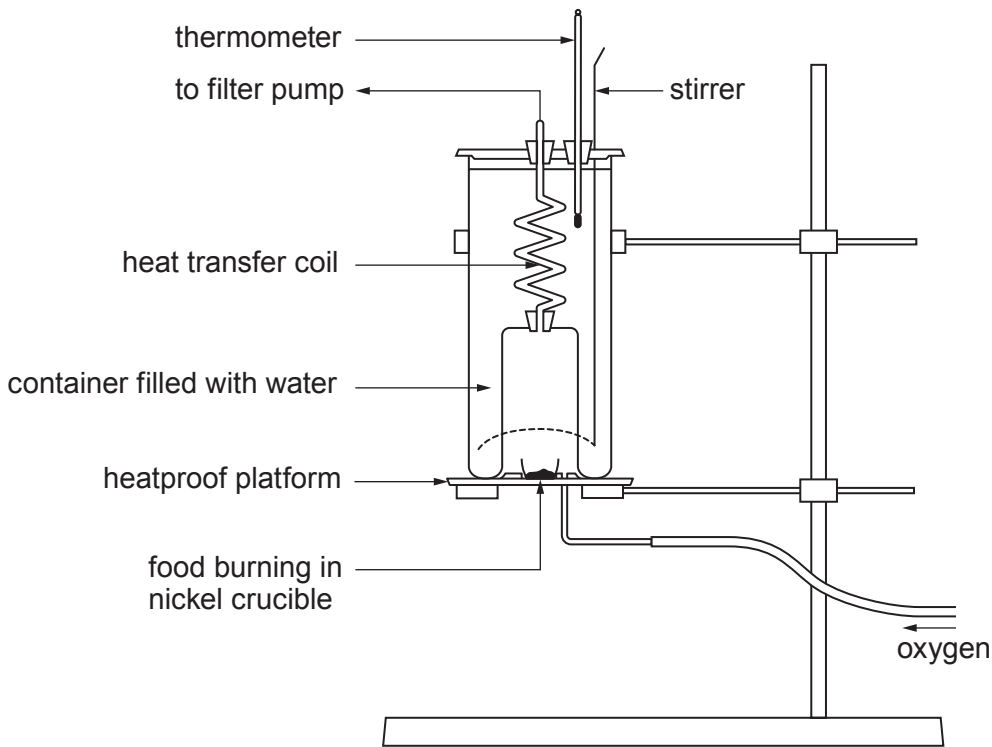
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- (d) Use the information from **Table 2** to calculate the percentage by which the value on the label for the crisps was greater than the value obtained in the experiment. [2]

Answer = %



- (e) The value calculated on the food label is worked out experimentally in a laboratory using apparatus similar to that shown below.



Suggest **two** ways in which the design of this apparatus makes it more efficient than the class experiment. [2]

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